
Generation of Believable Fake Combinational Logic Circuits for Cyber Deception

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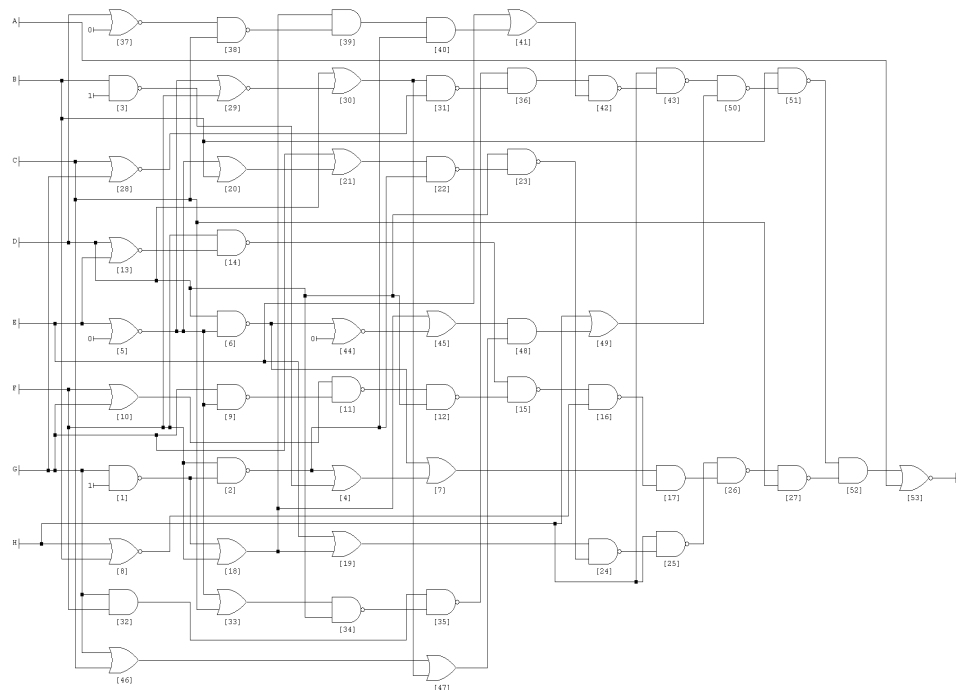
Abstract: The exponential increase in intellectual property (IP) theft has raised concern for organizations and government entities. Existing research suggests the use of fake document for protection against IP theft. To protect the logic circuit present in the IP document, we propose Believable Fake Combinational Logic Circuits Generation Engine (BFCLCGE), which is an extension of our previous work (FBLCGE). We addressed the exhaustive search problem by modifying the original Boolean equation and introducing replacement constraints, thereby reducing overall time complexity. Additionally, we overcame the limitation of FBLCGE by incorporating ESPRESSO algorithm for efficient minimization of Boolean equations with a large number of input variables. To test the believability of generated fake Boolean circuits, we implemented a virtual display (sixteen-segment) and simulated the outputs. The proposed approach is efficient and generates indistinguishable fake circuits, which can be used to create believable fake documents for cyber deception against potential IP theft.

Keywords: Intellectual Property ; Data Ex-filtration ; Combinational Logic Circuits ; Hardware Security ; Cyber Security ; Cyber Deception ; Decoy Logic Circuits ; Decoy Documents

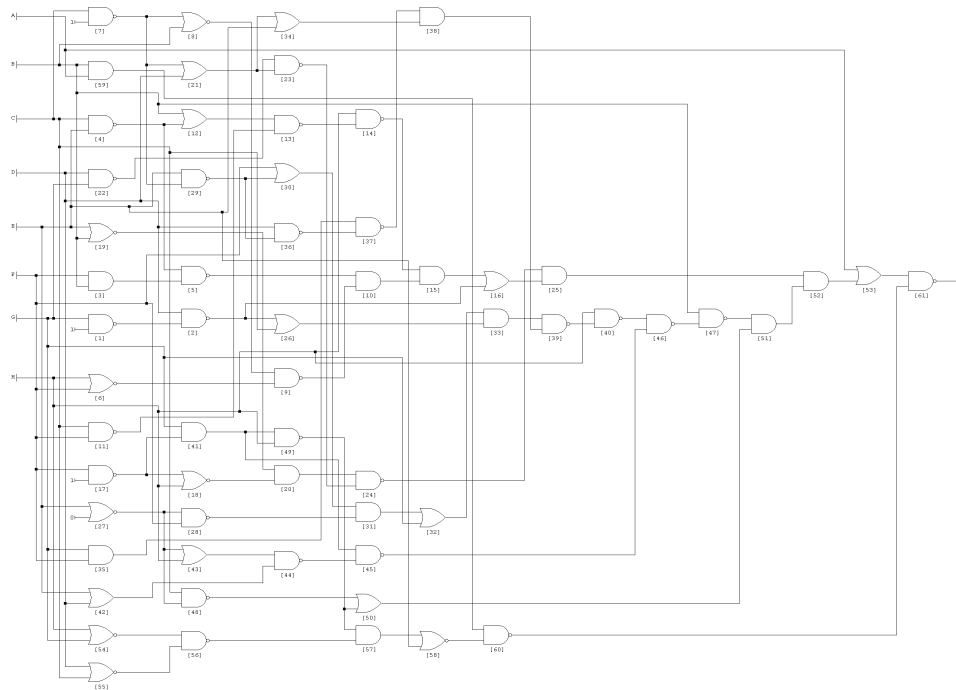
Biographical notes: **Nilin Prabhaker** received his B.Sc. (Computer Science) degree from Nilambar Pitambar University, Jharkhand, India. He earned his MCA from Indira Gandhi National Open University (IGNOU), New Delhi, India. He has received the Junior Research Fellow (UGC-JRF) award on 31st December 2019. He is currently pursuing his Ph.D. in Department of Computer Applications, National Institute of Technology, Tiruchirappalli, India. His research interests lie in the area of Cyber Security, Cyber Deception, Machine Learning, Deep Learning, Natural Language Processing, and Generative AI.

Ghanshyam S. Bopche Assistant Professor, Department of Computer Applications, National Institute of Technology, Tiruchirappalli. He received his MCA and B.Sc. (Electronics) degree from Nagpur University, Maharashtra, India. He earned his Ph.D. from Institute for Development and Research in Banking Technology (IDRBT) and University of Hyderabad, India. He was exchange research scholar at the State University of New York (SUNY) at Buffalo, NY, USA during 2015. His research interests lie in the area of Computer and Network Security, Cyber Deception, Information Assurance, and Digital Forensics.

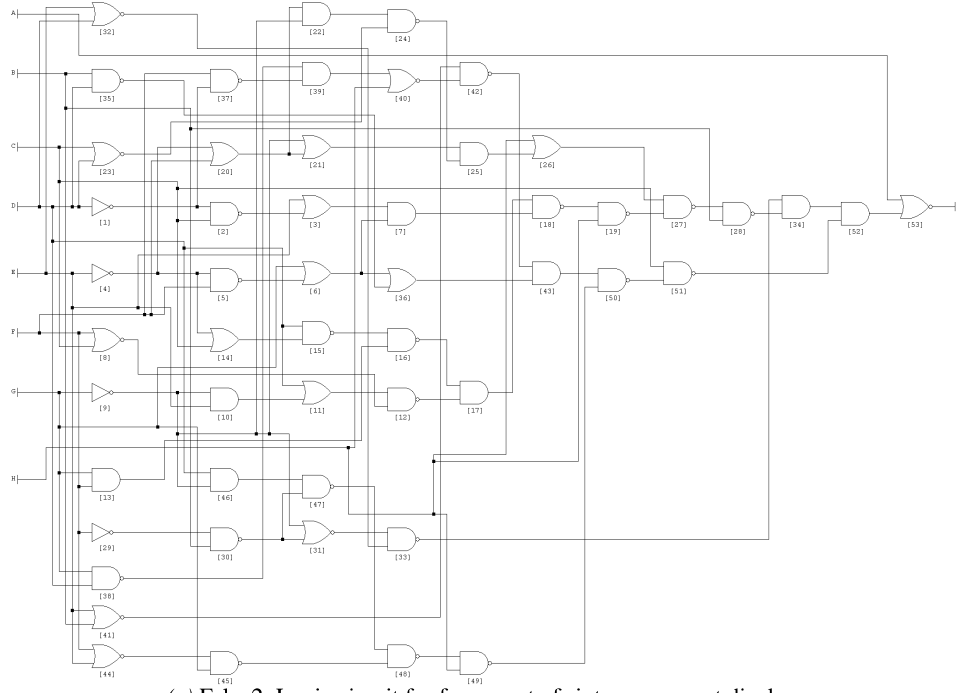
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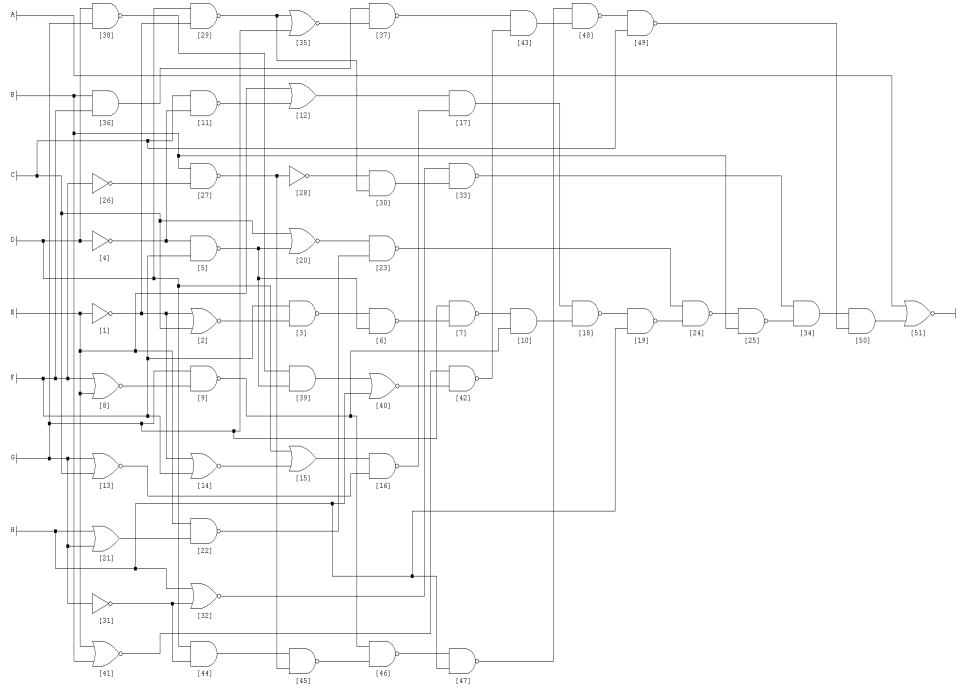
(a) Original: f-segment of sixteen-segment display



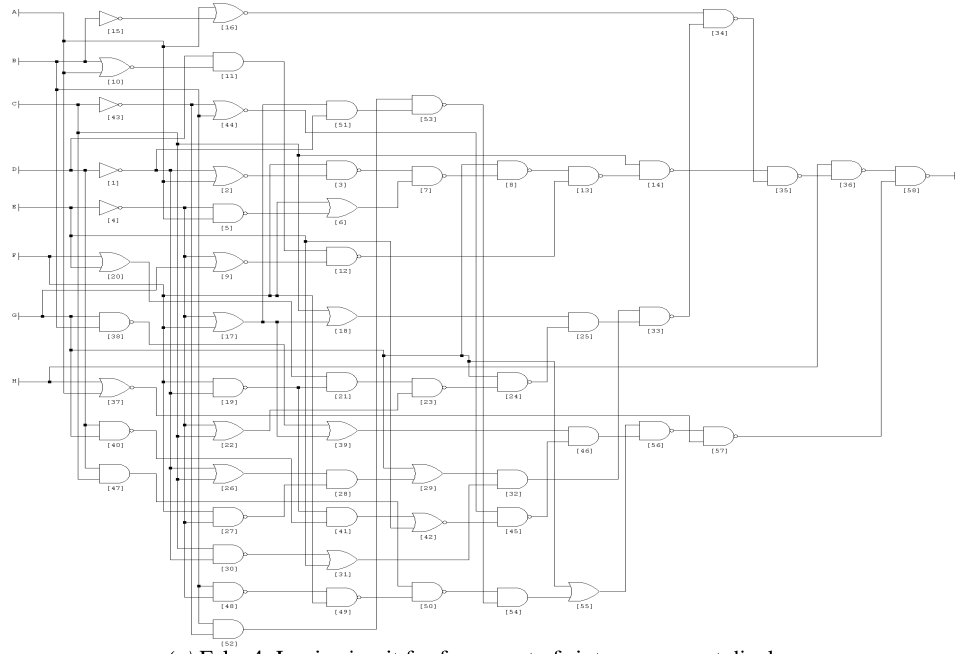
(b) Fake 1: Logic circuit for f-segment of sixteen-segment display



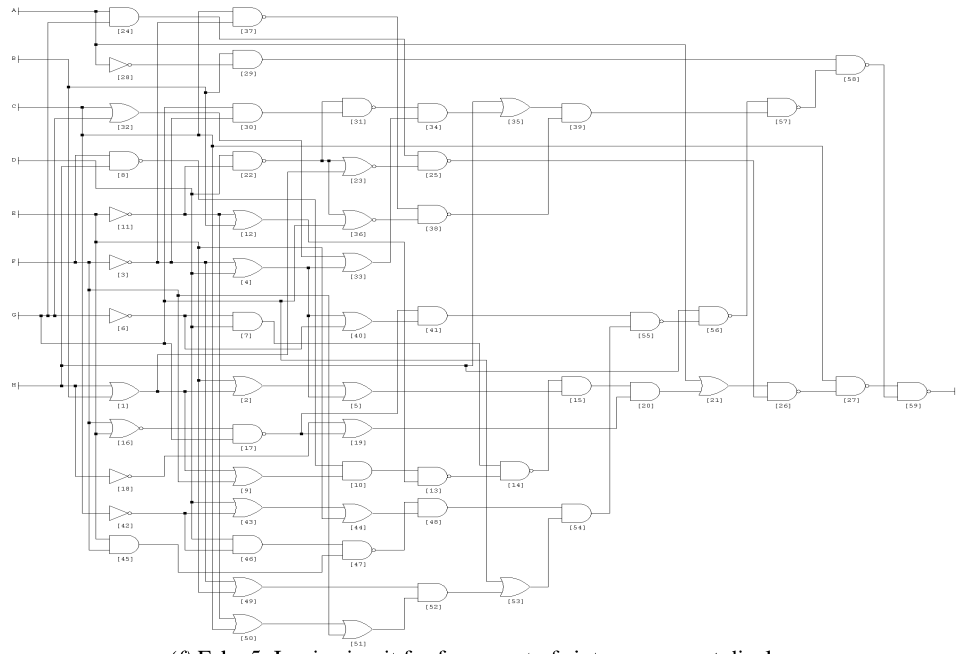
(c) Fake 2: Logic circuit for f-segment of sixteen-segment display



(d) Fake 3: Logic circuit for f-segment of sixteen-segment display

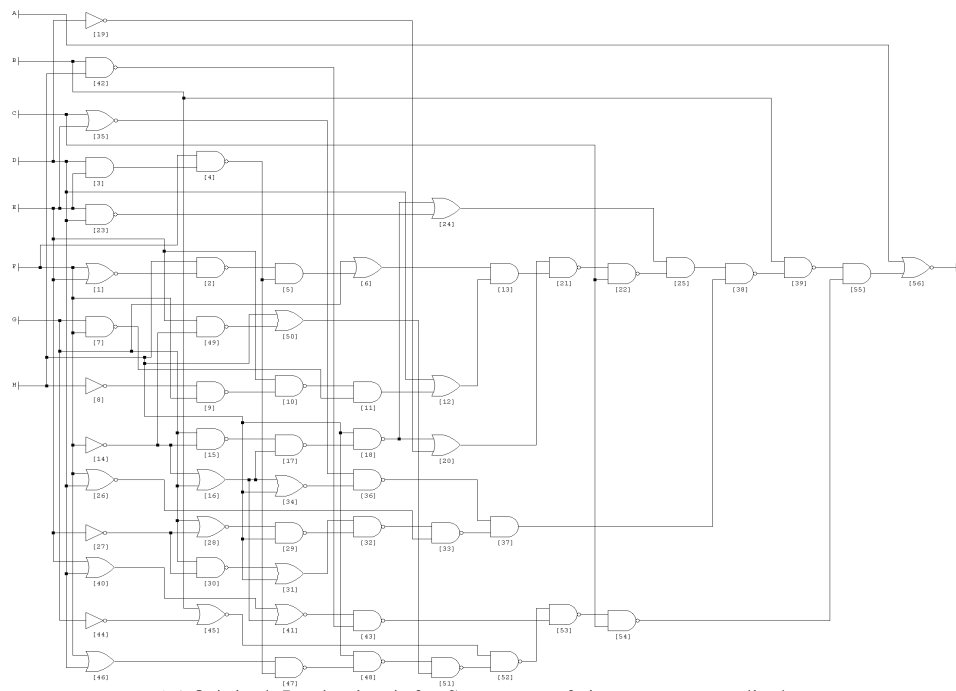


(e) Fake 4: Logic circuit for f-segment of sixteen-segment display

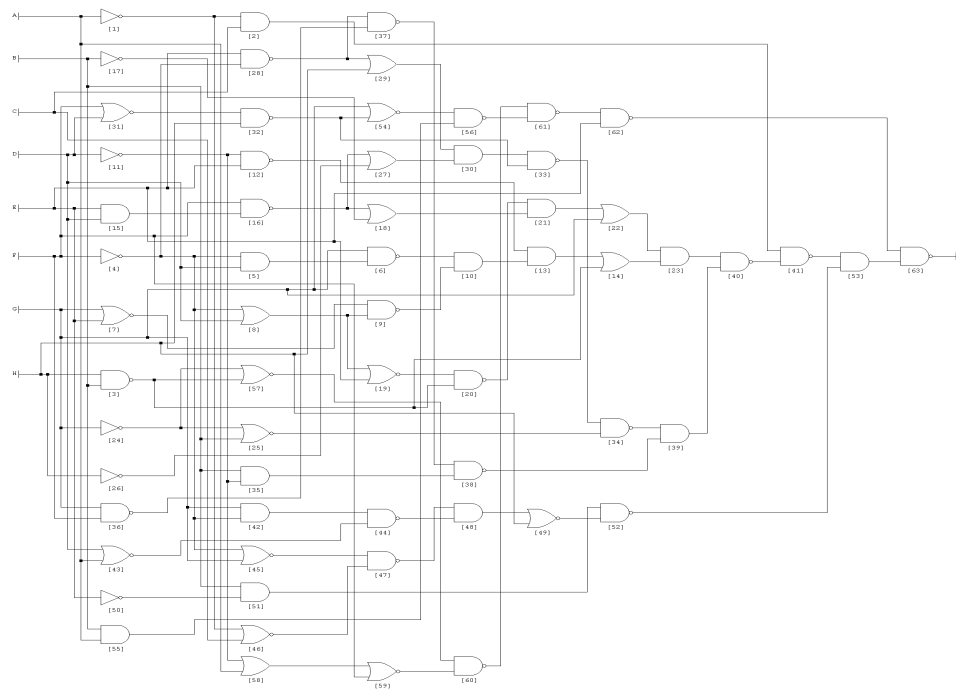


(f) Fake 5: Logic circuit for f-segment of sixteen-segment display

Figure 8: Original and Fake logic circuits for f-segment of sixteen-segment display. (a) represent the circuit diagram for original circuit. The generated fake version of original circuits are depicted in Sub-figures (b), (c), (d), (e), and (f).

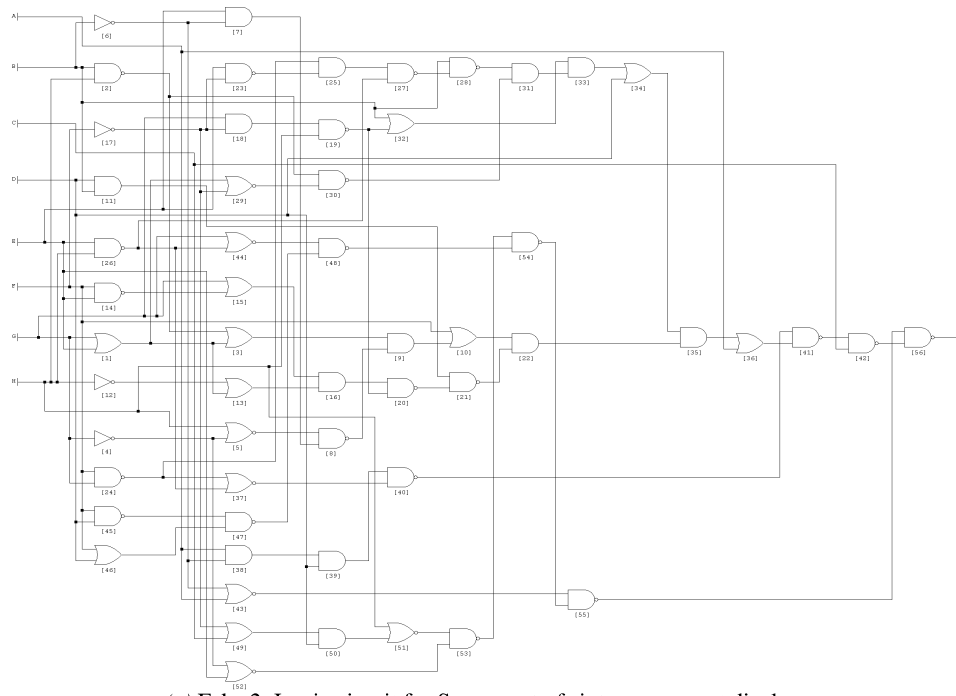


(a) Original: Logic circuit for S-segment of sixteen-segment display

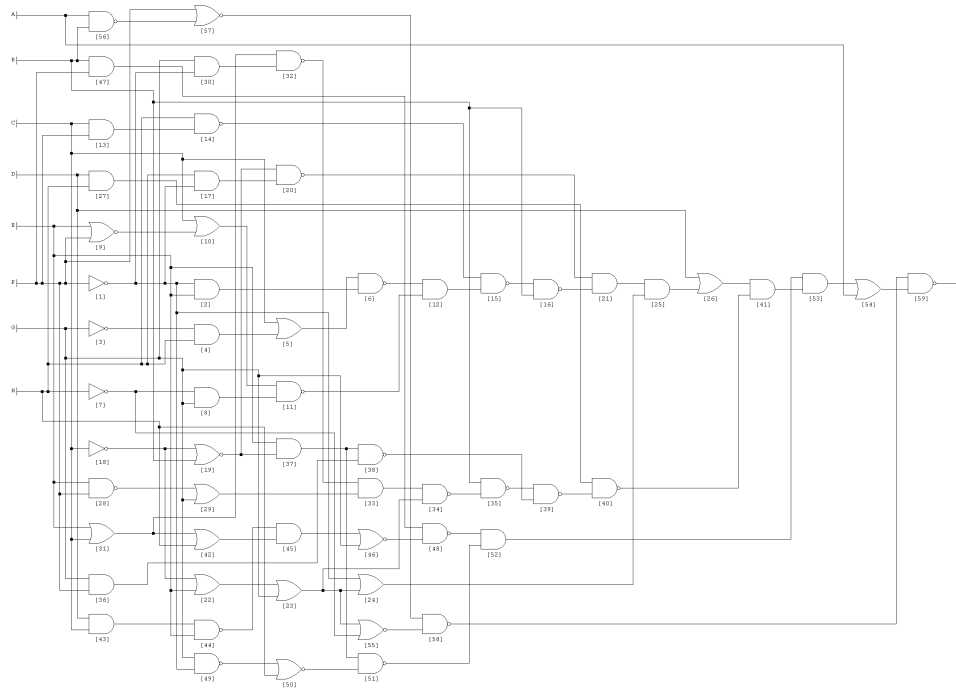


(b) Fake 1: Logic circuit for S-segment of sixteen-segment display

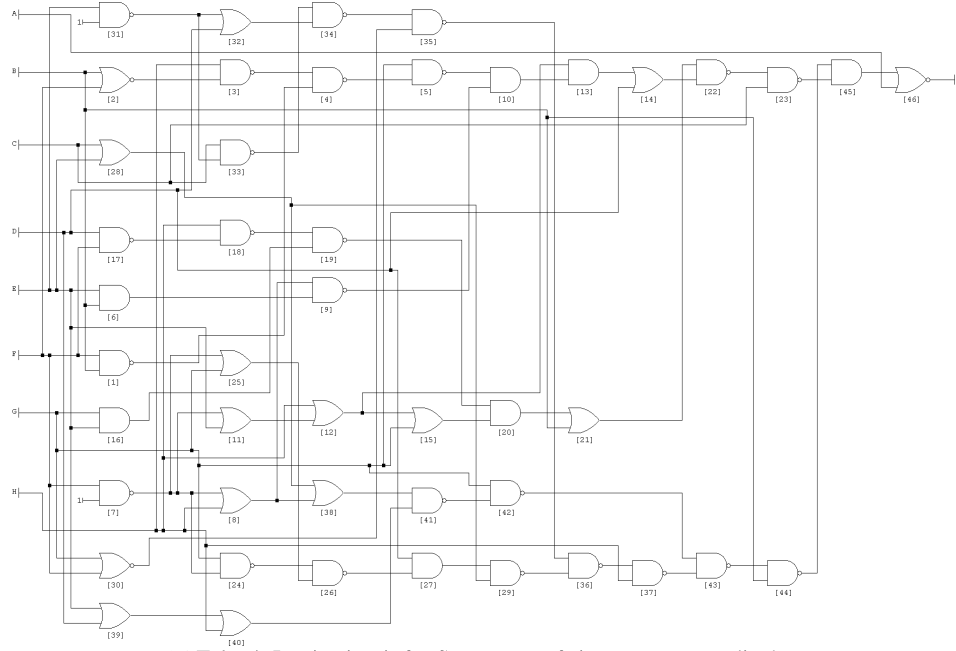
*Generation of Believable Fake Combinational Logic Circuits for Cyber Deception*²¹



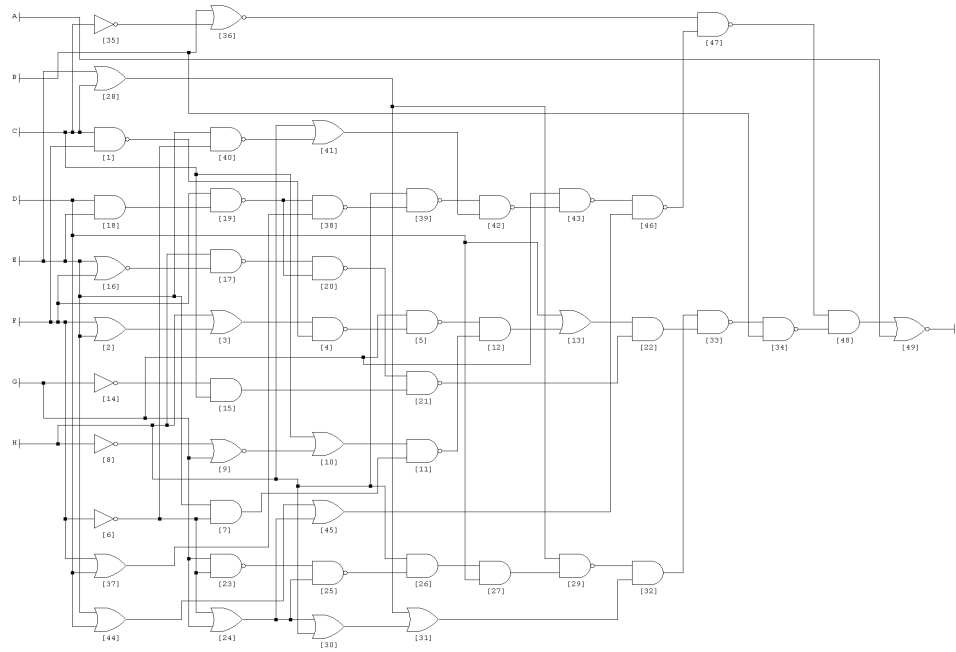
(c) Fake 2: Logic circuit for S-segment of sixteen-segment display



(d) Fake 3: Logic circuit for S-segment of sixteen-segment display

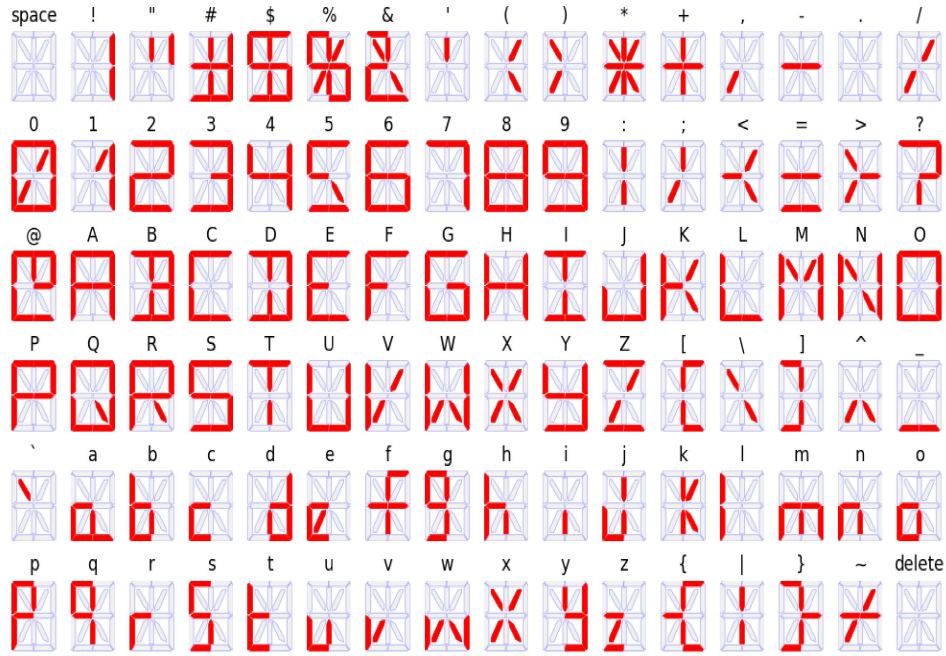


(e) Fake 4: Logic circuit for S-segment of sixteen-segment display

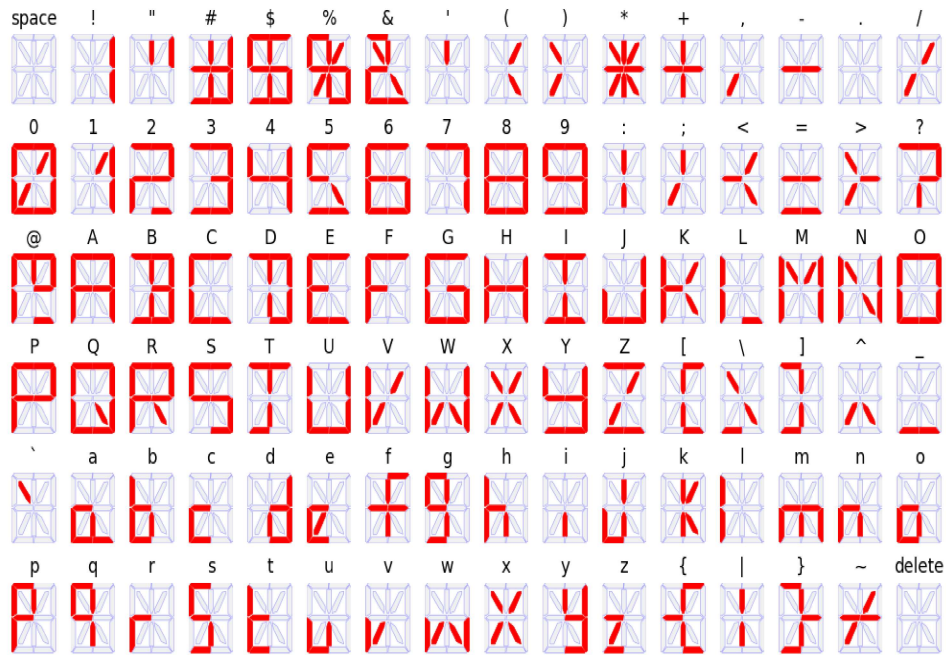


(f) Fake 5: Logic circuit for S-segment of sixteen-segment display

Figure 9: Original and Fake logic circuits for S-segment of sixteen-segment display. (a) represent the original circuit diagram, the generated fake version of original circuits are depicted in Sub-figures (b), (c), (d), (e), and (f)



(a) Original: Sixteen-segment display output (all characters, digits and symbols are represented with correct segments)

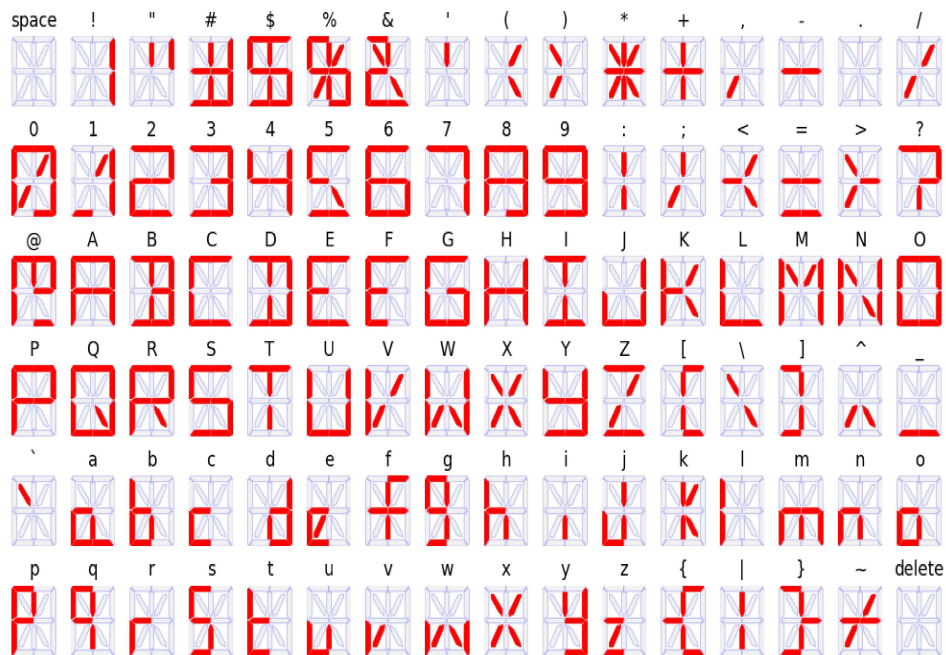


(b) Fake 1: output differs by a set of characters (D, L, and T), symbols (@, \), and digit (2).

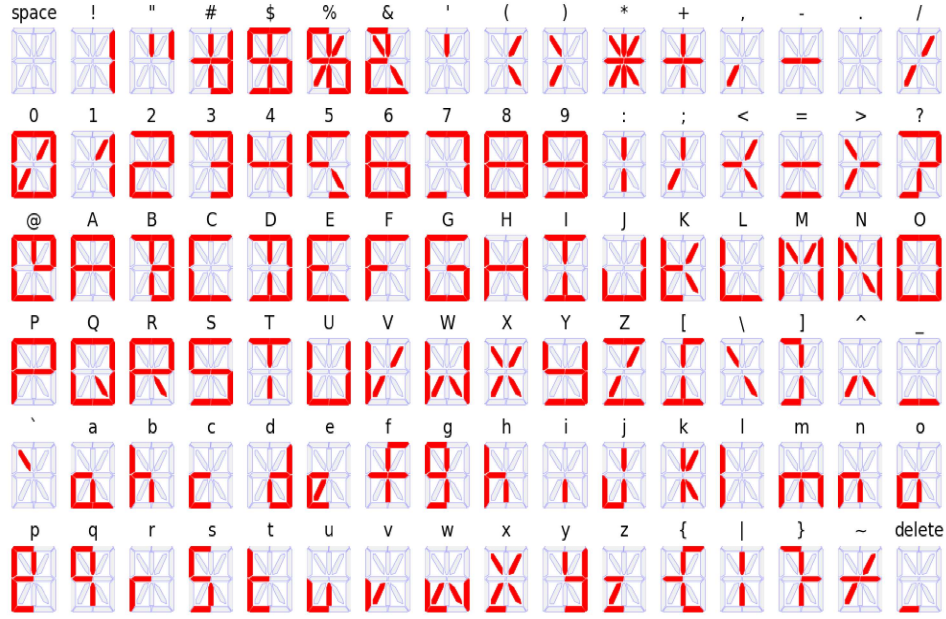
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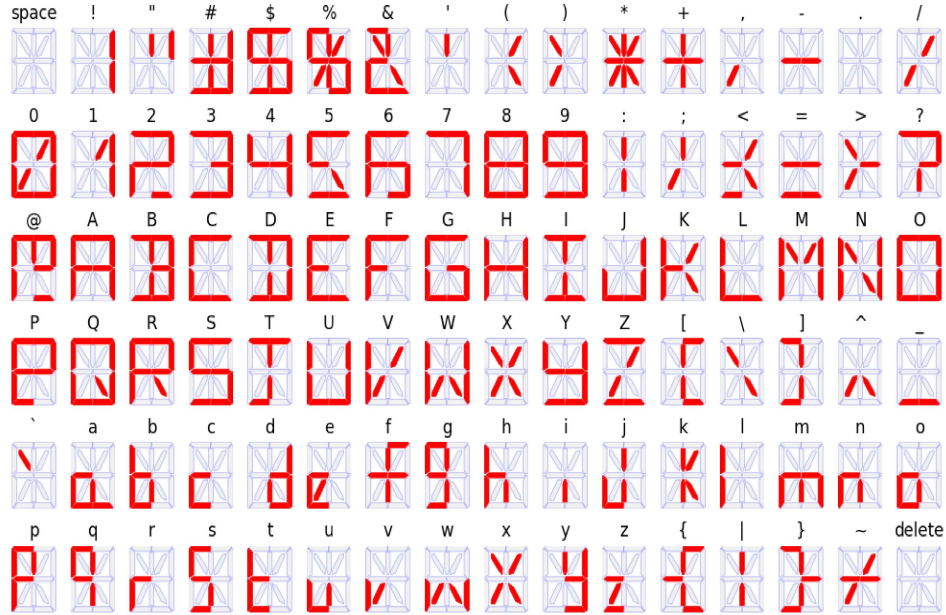
(c) Fake 2: output differs with the digits (0,1 and 8) and symbol (I).



(d) Fake 3: output differs with the digits (0,1 and 8) and symbol(@).

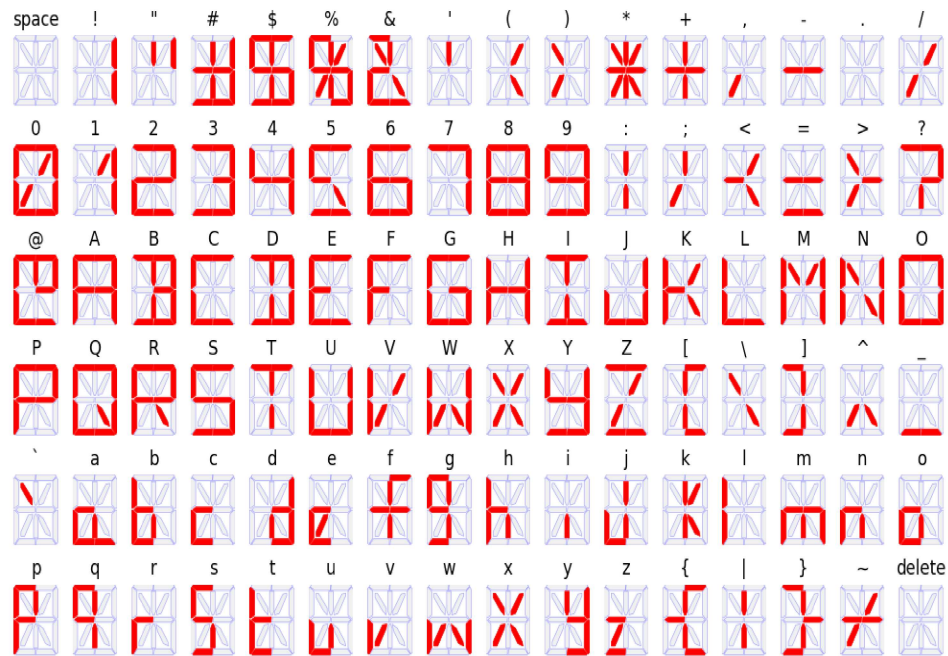


(e) Fake 4: output differs with digits 5 and 7, characters (B, K, b, w, and x), and special symbols (?, [, #, } and delete).



(f) Fake 5: output differs in output for digits (2 and 6), symbols(@ and <), and characters (P and T).

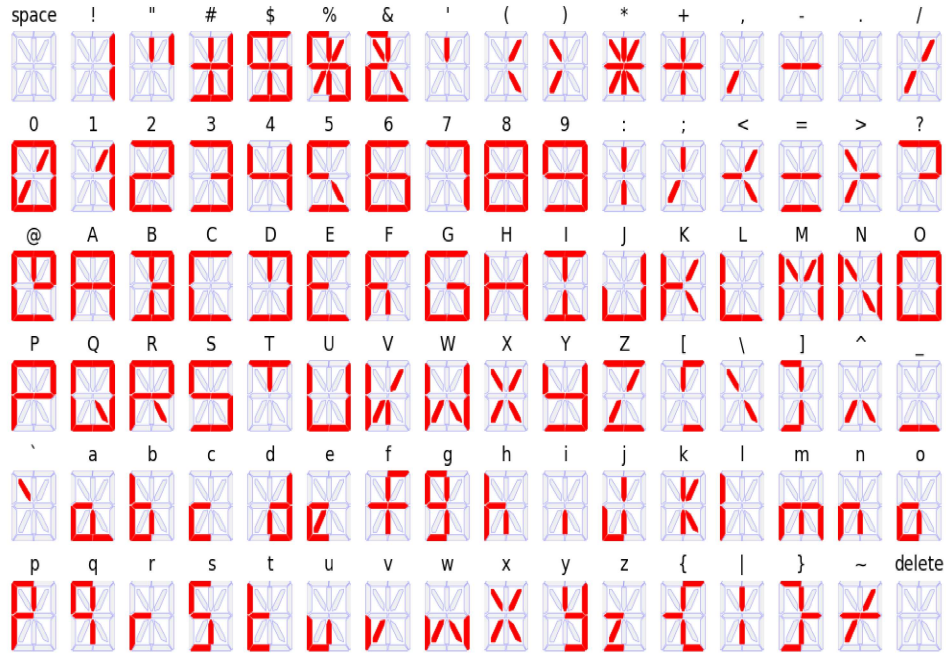
Figure 11: Sixteen-Segment Display Output for Original and Fake Logic Circuits after manipulation in f-segment: It is apparent that there is a significant difference in the output of fake logic circuits obtained through the experiment.



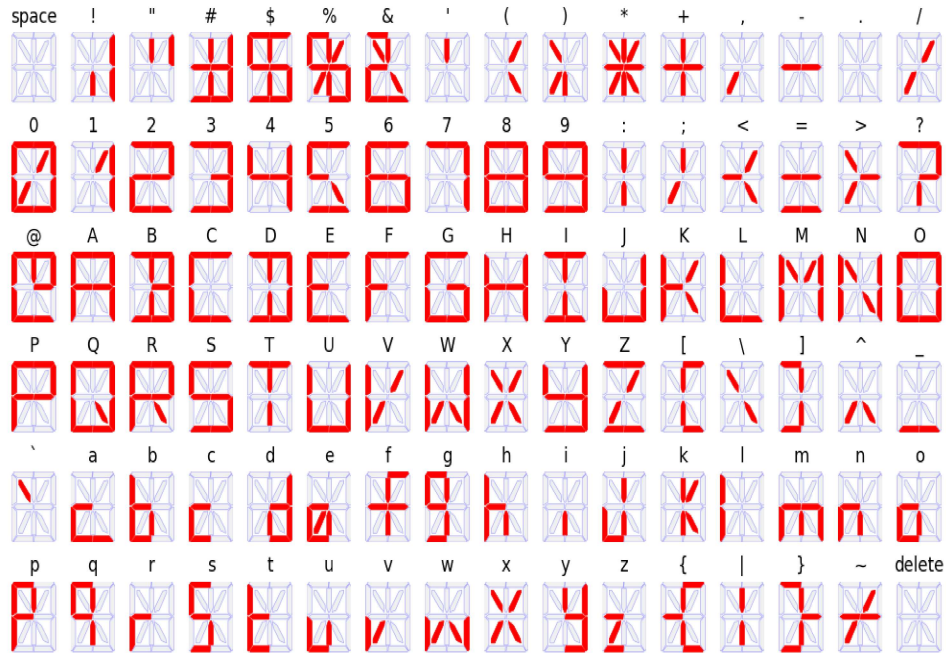
(a) Original: Sixteen-segment display output (all characters, digits and symbols are represented with correct segments)



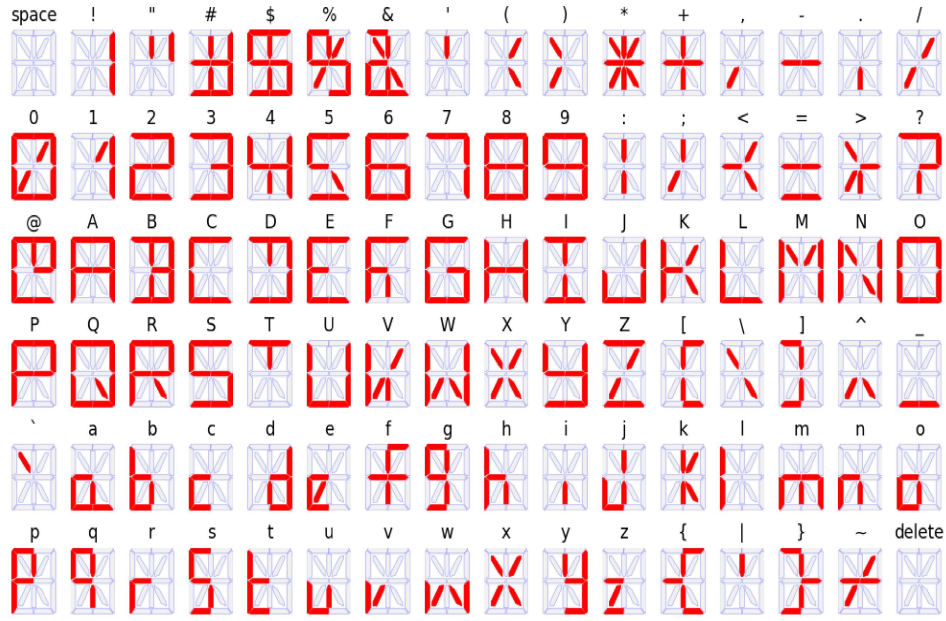
(b) Fake 1: incorrect output for the characters (D, I, and T) and symbol (j).



(c) Fake 2: incorrect outputs are ?, D, T, V, and [.



(d) Fake 3: output differs with symbol ")" and characters a, and e.



(e) Fake 4: output differs with &, 4, >, D, F, T, and V.



(f) Fake 5: output differs with character d and m.

Figure 12: Sixteen-Segment Display Output for Original and Fake Logic Circuits after manipulation in s-segment: It is evident that the output of display for fake circuits are significantly different from the output of the display obtained using Original circuit.

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(a) Message displayed using original circuits

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(c) Message displayed using Fake circuit 2

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(e) Message displayed using Fake circuit 4

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(b) Message displayed using Fake circuit 1

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(d) Message displayed using Fake circuit 3

```
ENGINE IGNITION FAILURE
ABORT MISSION IGNITION
TEMPRATURE LEVEL: 1500 C
ENGINE THRUST LEVEL: 0 KN
FUEL LEAK DETECTED
OXYGEN TANK PRESSURE LOW
CHECK LIFE SUPPORT SERVICE
FUEL TANK LEVEL: 10%
COMMUNICATIONS LINK LOST
SWITCH TO BACKUP MODE
NETW SIGNAL STRENGTH: 0 dB
```

(f) Message displayed using Fake circuit 5

Figure 14: Generated alert messages using concatenated original and fake sixteen-segment displays after manipulation in f-segment (the highlighted characters represent segments where the fake circuit introduces subtle structural modifications, resulting in distorted characters digits or symbols).

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(a) Message displayed using original circuits

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(c) Message displayed using Fake circuit 2

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(e) Message displayed using Fake circuit 4

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(b) Message displayed using Fake circuit 1

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(d) Message displayed using Fake circuit 3

```

LAST SIGNAL: 10 MINS AGO
VOLTAGE REACHED: 18.5 V
BATTERY LEVEL CRITICAL: 20%
BLOOD GLUCOSE CRITICAL
TRANSFORMER T3 OVERHEAT
CURRENT TEMPR OF T3: 98 C
VOLTAGE DROP AT NODE A3-184V
RADAR SCAN DELAY: 12.8 SEC
TRAIN 5421768 DELAYED
ESTIMATED ARRIVAL 18 MIN
MAINTENANCE WINDOW - TRACK 4
    
```

(f) Message displayed using Fake circuit 5

Figure 15: Generated alert messages using concatenated original and fake sixteen-segment displays after manipulation in s-segment (the highlighted characters represent segments where the fake circuit introduces subtle structural modifications, resulting in distorted characters digits or symbols).